

Math 170S - Homework 2 - due January 23

Problem 1. For the data from the problem 6.2-8, answer the following questions:

- (a) Find sample mean and sample standard deviation of the data.
- (b) Using the empirical rule, locate an interval that would contain approximately 95% of all data
- (c) Find the five-number summary of the data and draw a box plot
- (d) Find the 90th percentile

Problem 2. Let $Y_1 < Y_2 < \dots < Y_5$ be the ordered statistics of five independent observations from an exponential distribution that has a mean $\theta = 3$. Compute the probabilities $\mathbb{P}(Y_4 < 5)$ and $\mathbb{P}(1 < Y_1)$.

[Do not use the general formulas for pdf of order statistics (that you might find in the textbook). Instead, you can derive the probabilities in question from the definition of order statistics (like in Exercise 6.3-5(a) that we did in class. See also a similar Example 6.3-2).]

Problem 3. 6.3-14

Problem 4. 6.4-1 and 6.4-2.

What is similar and what is crucially different comparing to the example that we did in class (also, Example 6.4-3 in the red book, Example 6.4-5 in the blue book), when we estimated two unknown parameters of normal distribution $N(\theta_1, \theta_2)$?

Problem 5. 6.4 - 7

Problem 6. 6.4-17 (b) and (c) only